

Differentiation Worksheet — Chain Rule (Medium to Hard)

Nine Education IIT Academy | Class 11

Section A (Q1-20)

1. $y = \sin(5x^3 - 2x)$
2. $y = e^{(x^2 + 3x)}$
3. $y = (4x^2 - 7x + 1)^6$
4. $y = \ln(\tan x)$
5. $y = \sin^2(e^{(3x)})$
6. $y = \cos(3x^2 - 5x)$
7. $y = (3x - 5)^8$
8. $y = e^{(3x^2 - x)}$
9. $y = \ln(\sin x)$
10. $y = \sqrt{(3x^2 + 5x - 1)}$
11. $y = \tan(4x^2 + x)$
12. $y = (x^2 + 2x - 1)^5$
13. $y = \cos^2(3x)$
14. $y = e^{(\cos 2x)}$
15. $y = \ln(x^2 + 3x + 1)$
16. $y = \sin(\cos x)$
17. $y = \sec^2(5x)$
18. $y = (2x^3 - x)^4$
19. $y = e^{(\tan x)}$
20. $y = \cos(e^{(2x)})$

Section B (Q21-50) — Hard / Multi-Layer Chain Rule

21. $y = \sin^3(\cos 2x)$
22. $y = e^{(\sin^2 3x)}$
23. $y = \ln(\tan(x^2 + 1))$

- 24. $y = \sqrt{(\cos(e^{(2x)}))}$
- 25. $y = \tan(\ln(\sin x^3))$
- 26. $y = \cos^4(\ln 2x)$
- 27. $y = e^{(\cos^2(\sin x))}$
- 28. $y = \ln(\cos^2(3x^2 + 1))$
- 29. $y = \sin(e^{(\cos x)})$
- 30. $y = [\sin(3x^2 - x)]^5$
- 31. $y = \tan^2(e^{(2x)} + 1)$
- 32. $y = \ln[\sec(x^2) + \tan(x^2)]$
- 33. $y = \sqrt{(x^2 + \sin^2 3x)}$
- 34. $y = \sin(x^2) \cdot \cos(x^3)$
- 35. $y = x^2 \cdot e^{(\sin 3x)}$
- 36. $y = (e^{(2x)} + 1) / (e^{(2x)} - 1)$
- 37. $y = \ln[(x^2 + 1)/(x^2 - 1)]$
- 38. $y = \sin(2x) \cdot \cos(3x^2)$
- 39. $y = e^{(x^2)} \cdot \sin(e^{(x^2)})$
- 40. $y = (1 + \cos 2x)^3 / x$
- 41. $y = \ln(x + \sqrt{(x^2 + 1)})$
- 42. $y = \sin x \cdot e^{(\cos x)}$
- 43. $y = (x^3 + \ln x)^5$
- 44. $y = \cos(\sin(\tan x))$
- 45. $y = e^{(e^x)}$
- 46. $y = \ln(\ln x)$
- 47. $y = \sin^2 x \cdot \cos^2 x$
- 48. $y = \sqrt{(1 + \sin^2 2x)}$
- 49. $y = (e^x - e^{(-x)}) / (e^x + e^{(-x)})$
- 50. $y = \sin(x^2) \cdot \ln(\cos x) + x^3 e^{(2x)}$

Answer Key

Differentiation Worksheet — Chain Rule

Section A (Q1-20)

1. $\cos(5x^3 - 2x) \cdot (15x^2 - 2)$
2. $e^{(x^2 + 3x)} \cdot (2x + 3)$
3. $6(4x^2 - 7x + 1)^5 \cdot (8x - 7)$
4. $\sec^2 x / \tan x$
5. $3e^{(3x)} \cdot \sin(2e^{(3x)})$
6. $-\sin(3x^2 - 5x) \cdot (6x - 5)$
7. $24(3x - 5)^7$
8. $e^{(3x^2 - x)} \cdot (6x - 1)$
9. $\cot x$
10. $(6x + 5) / [2\sqrt{(3x^2 + 5x - 1)}]$
11. $\sec^2(4x^2 + x) \cdot (8x + 1)$
12. $10(x + 1)(x^2 + 2x - 1)^4$
13. $-3 \sin 6x$
14. $-2 \sin 2x \cdot e^{(\cos 2x)}$
15. $(2x + 3) / (x^2 + 3x + 1)$
16. $-\sin x \cdot \cos(\cos x)$
17. $10 \sec^2 5x \cdot \tan 5x$
18. $4(2x^3 - x)^3 \cdot (6x^2 - 1)$
19. $e^{(\tan x)} \cdot \sec^2 x$
20. $-2e^{(2x)} \cdot \sin(e^{(2x)})$

Section B (Q21-50)

21. $-6 \sin 2x \cdot \sin^2(\cos 2x) \cdot \cos(\cos 2x)$
22. $3 \sin 6x \cdot e^{(\sin^2 3x)}$
23. $2x \sec^2(x^2 + 1) / \tan(x^2 + 1)$
24. $-e^{(2x)} \sin(e^{(2x)}) / \sqrt{(\cos(e^{(2x)}))}$

25. $3x^2 \cot(x^3) \cdot \sec^2[\ln(\sin x^3)]$
26. $-(4/x) \cdot \sin(\ln 2x) \cdot \cos^3(\ln 2x)$
27. $-\cos x \cdot \sin(2 \sin x) \cdot e^{(\cos^2(\sin x))}$
28. $-12x \tan(3x^2 + 1)$
29. $-\sin x \cdot e^{(\cos x)} \cdot \cos(e^{(\cos x)})$
30. $5[\sin(3x^2 - x)]^4 \cdot \cos(3x^2 - x) \cdot (6x - 1)$
31. $4e^{(2x)} \cdot \tan(e^{(2x)} + 1) \cdot \sec^2(e^{(2x)} + 1)$
32. $2x \sec(x^2)$
33. $(2x + 3 \sin 6x) / [2\sqrt{(x^2 + \sin^2 3x)}]$
34. $2x \cos(x^2)\cos(x^3) - 3x^2 \sin(x^2)\sin(x^3)$
35. $e^{(\sin 3x)}[2x + 3x^2 \cos 3x]$
36. $-4e^{(2x)} / (e^{(2x)} - 1)^2$
37. $-4x / (x^4 - 1)$
38. $2\cos 2x \cdot \cos 3x^2 - 6x \sin 2x \cdot \sin 3x^2$
39. $2x e^{(x^2)}[\sin(e^{(x^2)}) + e^{(x^2)} \cos(e^{(x^2)})]$
40. $(1 + \cos 2x)^2[-6x \sin 2x - (1 + \cos 2x)] / x^2$
41. $1 / \sqrt{(x^2 + 1)}$
42. $e^{(\cos x)} \cdot [\cos x - \sin^2 x]$
43. $5(x^3 + \ln x)^4 \cdot (3x^2 + 1/x)$
44. $-\sin(\sin(\tan x)) \cdot \cos(\tan x) \cdot \sec^2 x$
45. $e^{(e^x)} \cdot e^x$
46. $1 / (x \ln x)$
47. $(1/2) \sin 4x$
48. $\sin 4x / \sqrt{(1 + \sin^2 2x)}$
49. $4 / (e^x + e^{-x})^2$
50. $2x \cos(x^2)\ln(\cos x) - \sin(x^2)\tan x + e^{(2x)}(3x^2 + 2x^3)$